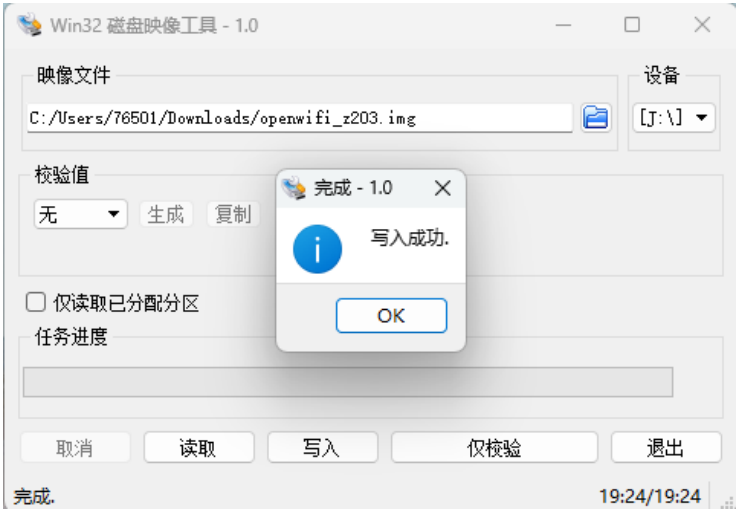


前提：将 openwifi-z203 镜像文件烧录到 SD 卡，SD 卡插入到设备，从 SD 卡启动设备，2 根 USB 线缆插入到 PC，网线插入到 PC，PC IP 地址更改为 192.168.10.1。



可将串口连接到 PC 串口终端，查看启动状态。

```
10. COM15 (USB Serial Port (COM15)) x
[ OK ] Started System Logging Service.
[ OK ] Started DHCP Client Daemon.
[ OK ] Finished dphys-swapfile - ...mount, and delete a swap file.
[ OK ] Started WPA supplicant.
[ OK ] Started LSB: Switch to on... (unless shift key is pressed).
[ OK ] Started LSB: rng-tools (Debian variant).
[ OK ] Started Avahi mDNS/DNS-SD Stack.
[ OK ] Started Authorization Manager.
[ OK ] Starting Modem Manager...
[ OK ] Started User Login Management.
[ OK ] Finished Raise network interfaces.
[ OK ] Reached target Network.
[ OK ] Reached target Network is Online.
[ OK ] Starting CUPS Scheduler...
[ OK ] Started Erlang Port Mapper Daemon.
[ OK ] Starting HTTP based time synchronization tool...
[ OK ] Starting Internet superserver...
[ OK ] Starting LSB: DHCP server...
[ OK ] Starting /etc/rc.local Compatibility...
[ OK ] Starting OpenBSD Secure Shell server...
[ OK ] Starting Permit User Sessions...
[ OK ] Started Unattended Upgrades Shutdown.
My IP address is 192.168.10.122
[ OK ] Starting LSB: Webfs simple HTTP server...
[ OK ] Started Internet superserver.
[ OK ] Finished Remove Stale Onli...ext4 Metadata Check Snapshots.
[ OK ] Started /etc/rc.local Compatibility.
[ OK ] Finished Permit User Sessions.
[ OK ] Started HTTP based time synchronization tool.
[ OK ] Started Modem Manager.
[ OK ] Started CUPS Scheduler.
[ OK ] Finished Analog Devices power up/down sequence.
[ OK ] Started Make remote CUPS printers available locally.
[ OK ] Starting Light Display Manager...
[ OK ] Starting Hold until boot process finishes up...
[ OK ] Started OpenBSD Secure Shell server.
[ OK ] Started Disk Manager.
[ OK ] Started LSB: Webfs simple HTTP server.
[ OK ] Finished Creating IIOD Context Attributes...
[ OK ] Started IIOD Daemon.
[FAILED] Failed to start VNC Server for X11.
[FAILED] Failed to start LSB: DHCP server.

Raspbian GNU/Linux 11 analog ttyPS0

analog login: root (automatic login)

Linux analog 5.15.36-27111-gc2f371e014f0-dirty #2 SMP PREEMPT Tue Aug 5 09:17:00 CEST 2025 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Aug 5 11:42:42 BST 2025 from 192.168.10.1 on pts/0
root@analog:~#
```

启动 Ubuntu 系统，打开终端。

第一步：

- Login to the board from your PC (PC Ethernet should have IP 192.168.10.1) with password openwifi.

```
ssh root@192.168.10.122
```

```
Activities Terminal 18:29 swd@swd: ~
File Edit View Search Terminal Help
swd@swd:~$ ssh root@192.168.10.122
root@192.168.10.122's password:
Linux analog 5.15.36-27111-gc2f371e014f0-dirty #2 SMP PREEMPT Tue Aug 5 09:17:00 CEST 2025 armv7l

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Last login: Tue Aug 5 11:42:23 2025
root@analog:~#
```

第二步:

`raspi-config --expand-rootfs` (Only needed when your SD card > 16GB. Run and reboot)

```
root@analog:~# raspi-config --expand-rootfs

Welcome to fdisk (util-linux 2.36.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): Disk /dev/mmcblk0: 29.12 GiB, 31266439168 bytes, 61067264 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x3c47d971

Device      Boot   Start      End  Sectors  Size Id Type
/dev/mmcblkp1 24576  4218879  4194304    2G c W95 FAT32 (LBA)
/dev/mmcblkp2 4218880 31116287 26897408  12.8G B3 Linux
/dev/mmcblkp3 8192    24575    16384    8M a2 unknown

Partition table entries are not in disk order.

Command (m for help): Partition number (1-3, default 3):
Partition 2 has been deleted.

Command (m for help): Partition type
   p  primary (2 primary, 0 extended, 2 free)
   e  extended (container for logical partitions)
Select (default p): Partition number (2,4, default 2): First sector (2048-61067263, default 2048): Last sector, +/-sectors or +/-size[K,M,G,T,P] (4218880-61067263, default 61067263)
:
Created a new partition 2 of type 'Linux' and of size 27.1 GiB.
Partition #2 contains a ext4 signature.

Command (m for help):
Disk /dev/mmcblk0: 29.12 GiB, 31266439168 bytes, 61067264 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x3c47d971

Device      Boot   Start      End  Sectors  Size Id Type
/dev/mmcblkp1 24576  4218879  4194304    2G c W95 FAT32 (LBA)
/dev/mmcblkp2 4218880 61067263 56848384  27.1G B3 Linux
/dev/mmcblkp3 8192    24575    16384    8M a2 unknown

Partition table entries are not in disk order.

Partition 2 has been deleted.

Command (m for help): Partition type
   p  primary (2 primary, 0 extended, 2 free)
   e  extended (container for logical partitions)
Select (default p): Partition number (2,4, default 2): First sector (2048-61067263, default 2048): Last sector, +/-sectors or +/-size[K,M,G,T,P] (4218880-61067263, default 61067263)
:
Created a new partition 2 of type 'Linux' and of size 27.1 GiB.
Partition #2 contains a ext4 signature.

Command (m for help):
Disk /dev/mmcblk0: 29.12 GiB, 31266439168 bytes, 61067264 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x3c47d971

Device      Boot   Start      End  Sectors  Size Id Type
/dev/mmcblkp1 24576  4218879  4194304    2G c W95 FAT32 (LBA)
/dev/mmcblkp2 4218880 61067263 56848384  27.1G B3 Linux
/dev/mmcblkp3 8192    24575    16384    8M a2 unknown

Partition table entries are not in disk order.

Command (m for help): The partition table has been altered.
Syncing disks.

perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
    LANGUAGE = (unset),
    LC_ALL = (unset),
    LC_TIME = "zh_CN.UTF-8",
    LC_MONETARY = "zh_CN.UTF-8",
    LC_ADDRESS = "zh_CN.UTF-8",
    LC_TELEPHONE = "zh_CN.UTF-8",
    LC_NAME = "zh_CN.UTF-8",
    LC_MEASUREMENT = "zh_CN.UTF-8",
    LC_IDENTIFICATION = "zh_CN.UTF-8",
    LC_NUMERIC = "zh_CN.UTF-8",
    LC_PAPER = "zh_CN.UTF-8",
    LANG = "en_US.UTF-8"
are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("en_US.UTF-8").
Please reboot
root@analog:~#
```

第三步: 板卡断电重启, 重启完成后再次登录。

```
File Edit View Search Terminal Help
swd@swd:~$ ssh root@192.168.10.122
root@192.168.10.122's password:
Linux analog 5.15.36-27111-gc2f371e014f0-dirty #2 SMP PREEMPT Tue Aug 5 09:17:00 CEST 2025 armv7l

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permitted by applicable law.
Last login: Tue Aug 5 11:42:41 2025
root@analog:~#
```

第四步:

`./openwifi/setup_once.sh` (Reboot the board. Only need to run once for new board)

[illegible]

```
File Edit View Search Terminal Help
swd@swd:~$ ssh root@192.168.10.122
root@192.168.10.122's password:
Linux analog 5.15.36-27111-gc2f371e014f0-dirty #2 SMP PREEMPT Tue Aug 5 09:17:00 CEST 2025 armv7l

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permitted by applicable law.
Last login: Tue Aug 5 11:42:23 2025
root@analog:~#
```

```
cd openwifi
./wgd.sh
```

```

root@analog:~# cd openwifl
root@analog:~/openwifl# ./jgd.sh
usage:
  Script for load (or download+load) different driver and FPGA img without rebooting
  no argument: Load .ko driver files and FPGA img (if system_top.bit.bin exist) in current dir with test_mode=0.
  1st argument: If it is a NUMBER, it will be assigned to test_mode. Then load everything from current dir.
  1st argument: If it is a string called "remote", it will download driver/fpga and load everything.
  - 2nd argument (if exist) is the target directory name for downloading and reloading
  - 3rd argument (if exist) is the value for test_mode
  1st argument: neither NUMBER nor "remote" nor a .tar.gz file, it is regarded as a directory and load everything from it.
  - 2nd argument (if exist) is the value for test_mode
  1st argument: a .tar.gz file, it will be unpacked then load from that unpacked directory
  - 2nd argument (if exist) is the value for test_mode

Module      Size  Used by
xilinx_dna  28672  0
8021q       28672  0
TARGET_DIR ./
DOWNLOAD_FLAG 0
test_mode 0
tx_offset_tuning_enable 0
$TARGET_DIR is found!

hostapd: no process found
dhcpcd: no process found
sdr0: ERROR while getting interface flags: No such device
rmmod: ERROR: Module sdr is not currently loaded
./load_fpga_img.sh: line 14: warning: command substitution: ignored null byte in input
xlnx.zynq-7000
//system_top.bit.bin
+ ifconfig sdr0 down
sdr0: ERROR while getting interface flags: No such device
+ rmmod sdr
rmmod: ERROR: Module sdr is not currently loaded

```

```

fast_attack
40000000
25215513
40000000
25215414
rs1
112.25 dB
100.00 dB
rx0 gain to 70
65.000000 dB
./rf_intf_11n.sh: 101: echo: echo: I/O error
65.000000 dB
rx1 gain to 70
55.000000 dB
./rf_intf_11n.sh: 106: echo: echo: I/O error
55.000000 dB
tx0 gain -89dB
-10.000000 dB
-89.000000 dB
tx1 gain 0dB
-10.000000 dB
0.000000 dB
rmmod: ERROR: Module tx_intf is not currently loaded
+ sudo insmod ./tx_intf.ko
tx_intf is loaded!
rmmod: ERROR: Module rx_intf is not currently loaded
+ sudo insmod ./rx_intf.ko
rx_intf is loaded!
rmmod: ERROR: Module openofdm_tx is not currently loaded
+ sudo insmod ./openofdm_tx.ko
openofdm_tx is loaded!
rmmod: ERROR: Module openofdm_rx is not currently loaded
+ sudo insmod ./openofdm_rx.ko
openofdm_rx is loaded!
rmmod: ERROR: Module xpu is not currently loaded
+ sudo insmod ./xpu.ko
xpu is loaded!
rmmod: ERROR: Module sdr is not currently loaded
+ sudo insmod ./sdr.ko test_mode=0
sdr is loaded!
+ test -f /sys/kernel/debug/ltio/ltio:device0/direct_reg_access
+ cd /sys/kernel/debug/ltio/ltio:device0/
+ set +x
Applied optimized AGC settings
the end
root@analog:~/openwifl#

```

第七步:

```

./fosdem.sh

```

```

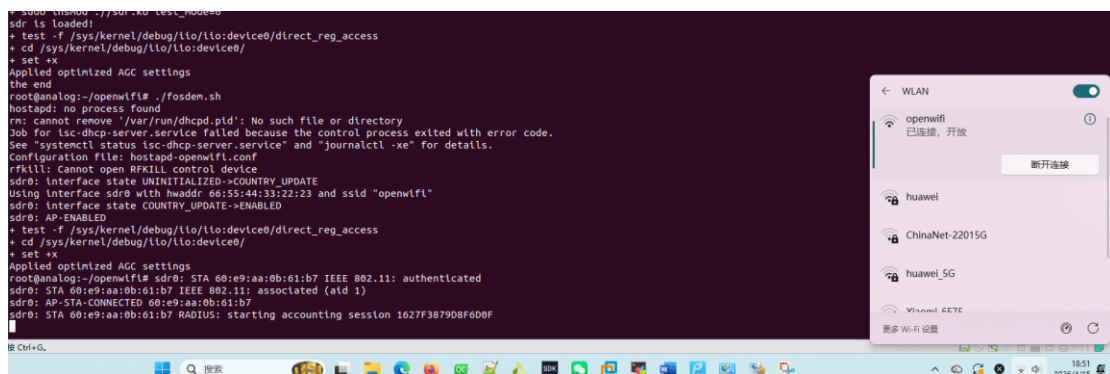
root@analog:~/openwifl# ./fosden.sh
hostapd: no process found
rm: cannot remove '/var/run/dhcpd.pid': No such file or directory
Job for isc-dhcp-server.service failed because the control process exited with error code.
See "systemctl status isc-dhcp-server.service" and "journalctl -xe" for details.
Configuration file: hostapd-openwifl.conf
rfkill: Cannot open RFKILL control device
sdr0: interface state UNINITIALIZED->COUNTRY UPDATE
Using interface sdr0 with hwaddr 66:55:44:33:22:d5 and ssid "openwifl"
sdr0: interface state COUNTRY_UPDATE->ENABLED
sdr0: AP-ENABLED
+ test -f /sys/kernel/debug/ltio/ltio:device0/direct_reg_access
+ cd /sys/kernel/debug/ltio/ltio:device0/
+ set +x
Applied optimized AGC settings
root@analog:~/openwifl#

```

第八步：然后在 PC 端连接 openwifi，如果连不上或扫描不到 openwifi 建议重启一次，重启后登录终端执行第六步和第七步指令。



连接 openwifi 后终端提示状态。



连接后访问 192.168.13.1，网页正常打开可播放视频，即证明 openwifi 启动正常。

